

Big Mountain Resort

Montana, United States
March 2026

Problem Identification

Big Mountain Resort which is a ski resort in state of Montana is interested to know how it can make more profit from their facilities.

The management team believes the resort's ticket does not match features and facilities available at the resort. Therefore a Data Science team is hired to investigate how the resort can achieve its goal.

At the very high level, the problem is divided in two parts:

- 1) How much the resort can increase ticket price?
- 2) How the resort management team can cut extra costs?

Key Findings

State, population of the state, and density of resorts in the state are not critical and do not play important role in this business.

Key features and facilities which are important for a ski resort to be competitive in the market are listed as below:

- Vertical Drop
- Snow Making
- Total lift/chairs
- Run
- Longest Run
- Skiable terrain area

Modeling & Analysis

Big Mountain Resort stays among top ski resorts in the united states.

Among all of the key facilities and features which a ski resort should have, Big Mountain Resort is in a good shape and stands on top of the lists.

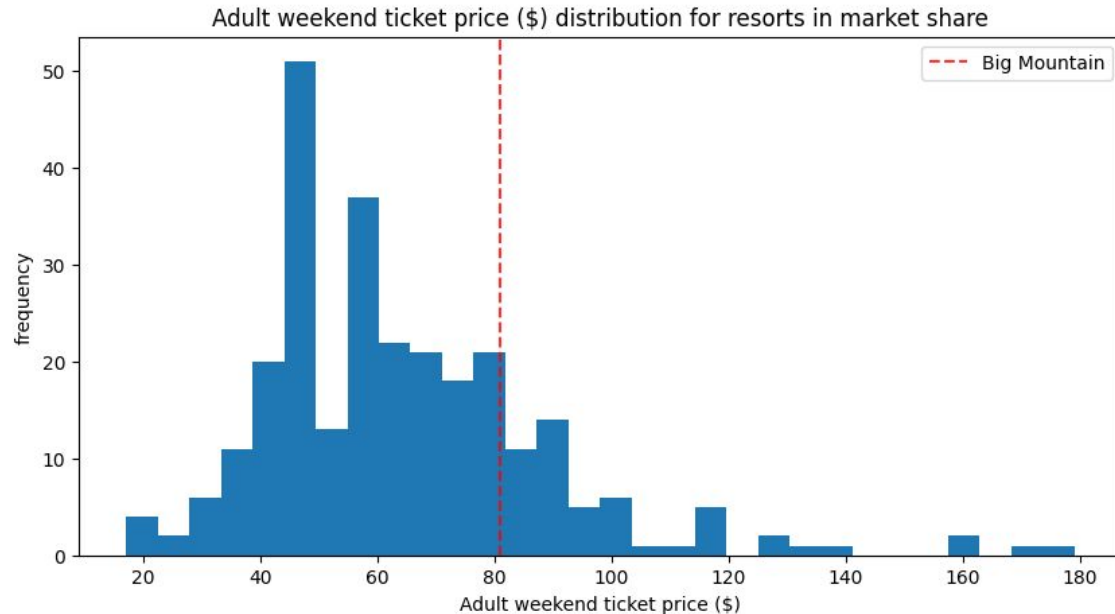
Current ticket price is \$81. However, the modelled price is \$95.87 which is about \$14 more than current ticket price.

Note: There is about \$10.39 margin error in the modelled price, \$95.87 .

Next few slides contain some graphs for clarification.

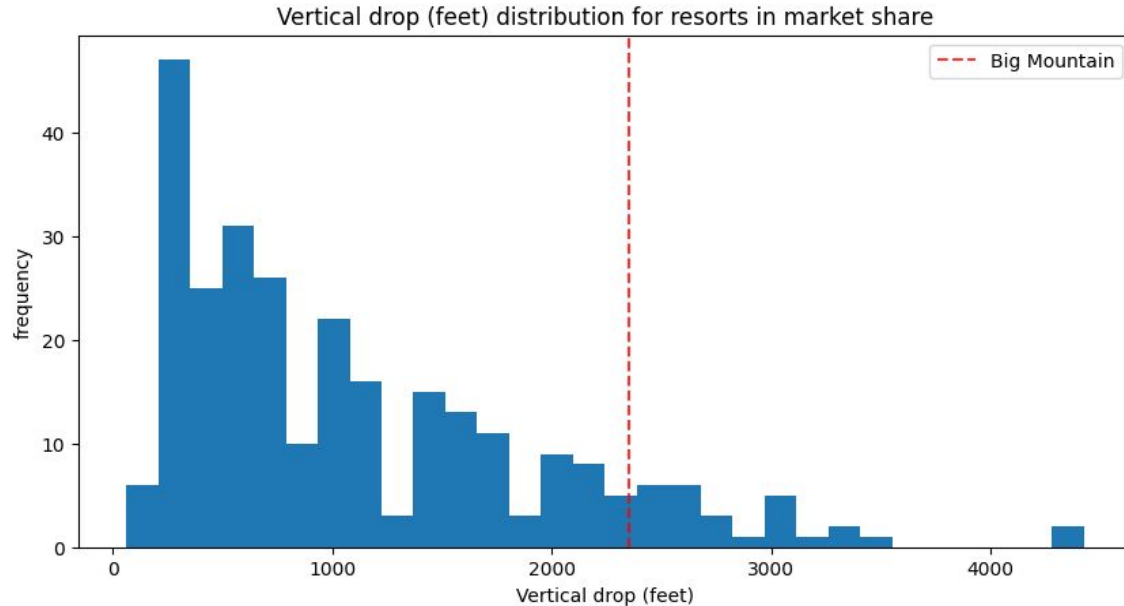
Modelling & Analysis _ ticket price

Big Mountain ticket price in the market



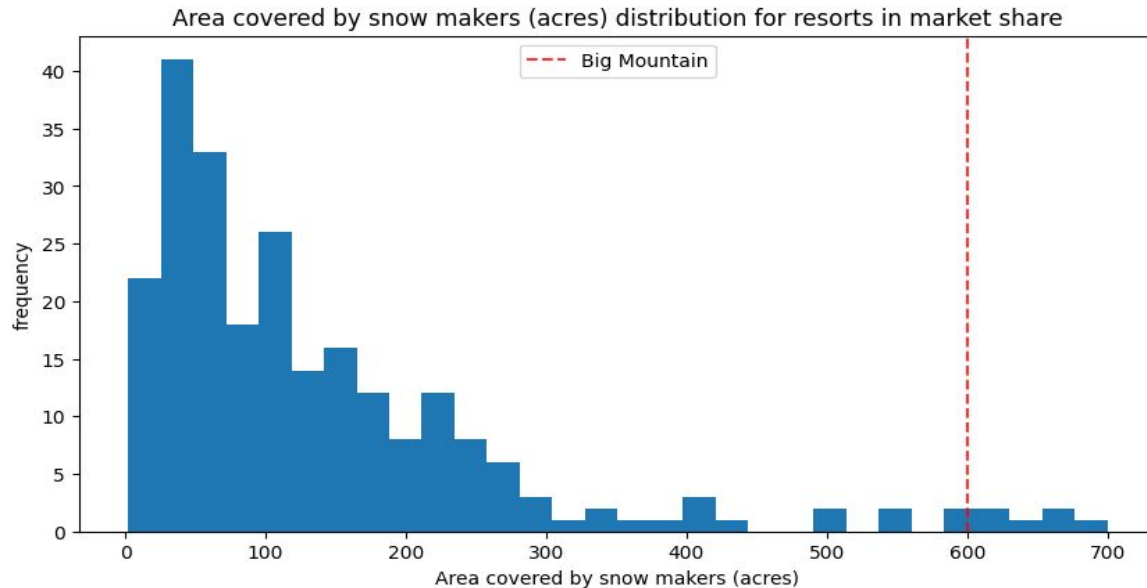
Modelling & Analysis _ vertical drop

Big Mountain vertical drop in the market



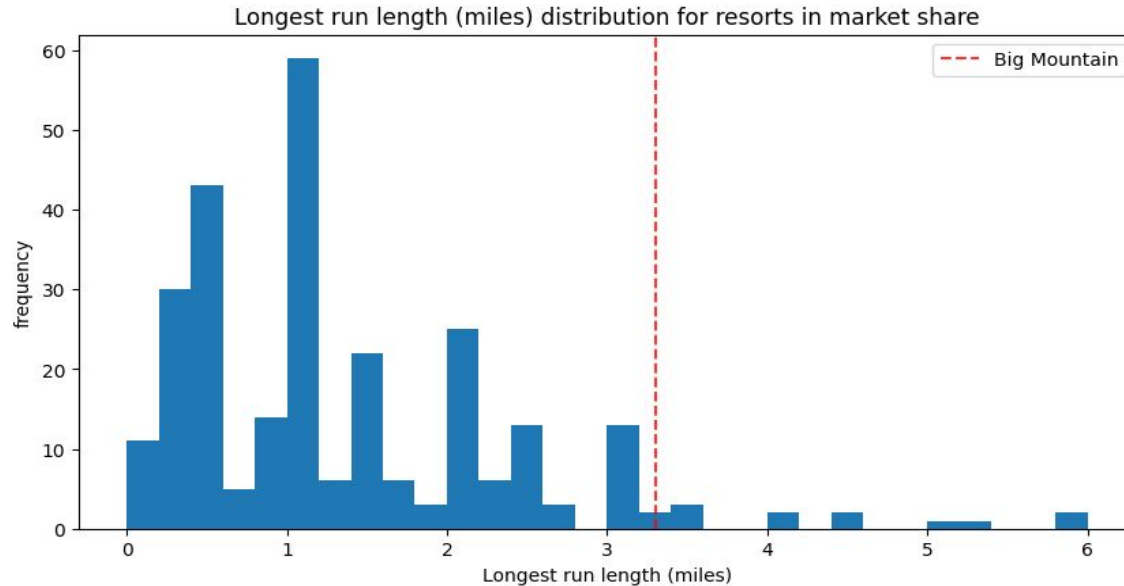
Modelling & Analysis _ snow making

Big Mountain snow making in the market



Modelling & Analysis _ longest run

Big Mountain longest run in the market



Recommendations

- 1) Add a Run and increase Vertical Drop which results in \$2 ticket price increment
- 2) If the business does not want to expand investment, then close one Run, because cutting one Run has no negative impact.
- 3) Consider margin of error and add ticket price by \$4.

Summary & Conclusion

Big Mountain is doing well in ski resort business market.

Having extensive features and facilities place Big Mountain resort among top ski resorts in the united states.

As mentioned in Recommendations slide, increasing ticket price and cutting extra costs are both valid options for Big Mountain to make more profit.